

1. Literature Survey

Flow-Based Intrusion Detection System for SDN

Paper:

Flow-Based Intrusion Detection System for SDN

Objective

The main objective of this research is to design a **lightweight flow-based Intrusion Detection System (IDS)** for **Software Defined Networks (SDN)**. The proposed system uses **statistical information collected from OpenFlow switches** to detect malicious network behavior. The approach aims to improve detection accuracy while reducing overhead in SDN networks.

Background

Software Defined Networking separates the **control plane** from the **data plane**, enabling centralized network management. While SDN improves network flexibility, it also introduces new vulnerabilities such as attacks on **northbound and southbound interfaces**. Traditional IDS approaches rely heavily on packet-level analysis, which increases computational cost and processing overhead.

Proposed Method

The proposed IDS architecture operates using **flow-based traffic analysis** rather than packet inspection. The system periodically collects flow statistics from OpenFlow switches and analyzes the traffic characteristics using machine learning classification techniques.

The system architecture consists of the following components:

1. Flow statistics collection from OpenFlow switches
2. Feature extraction from traffic flows
3. Aggregation of extracted features
4. Machine learning classification of network traffic

The flow statistics include metrics such as packet count, byte count, duration of flow, source and destination IP addresses, transport protocol type, and port numbers.

Feature Extraction

According to the **feature extraction table on page 3**, the system extracts key flow attributes such as:

- Duration of communication
- Packet count per flow
- Byte count per flow
- Source and destination IP addresses
- Transport protocol (TCP, UDP, ICMP)
- Source and destination port numbers

These features help characterize network traffic patterns and distinguish between normal and malicious flows.

System Workflow

The proposed IDS works through the following process:

1. When a new flow arrives at the OpenFlow switch, the controller collects statistics.
2. Flow statistics are recorded every second.
3. Extracted features are aggregated to represent traffic behavior.
4. A supervised machine learning model classifies the flow as **normal or malicious**.

The **flow-based detection architecture shown in the diagram on page 2** illustrates how traffic statistics are collected from switches and sent to the controller for classification.

Advantages

The proposed system offers several benefits:

- Reduced computational overhead compared to packet-based IDS
- Real-time traffic monitoring
- Transparent operation without requiring host information
- Scalable design for SDN environments

Experimental Results

The experimental results demonstrate that the system achieves:

- **F1-score of approximately 0.98**
- High detection rate
- Low false alarm rate

These results indicate that the flow-based IDS is effective in detecting attacks such as:

- Denial of Service (DoS)
- HTTP brute force attacks

- SSH brute force attacks.

Conclusion

The study concludes that leveraging **flow statistics in SDN environments** allows efficient intrusion detection without requiring packet-level inspection. The proposed IDS improves detection accuracy while maintaining low computational overhead, making it suitable for deployment in SDN networks.

2. Literature Survey

Ensemble Learning Approach for Flow-Based Intrusion Detection System

Paper:

Ensemble Learning Approach for Flow-Based Intrusion Detection System

Objective

This research proposes a **flow-based intrusion detection system using ensemble machine learning techniques**. The goal is to improve detection accuracy and reduce false alarms compared to traditional single-classifier IDS models.

Background

Network security threats continue to grow with the expansion of Information and Communication Technologies (ICT). Traditional IDS methods relying on **deep packet inspection** suffer from computational limitations and quality-of-service issues.

Flow-based IDS methods analyze **network flow records instead of full packets**, which significantly reduces processing requirements and improves scalability.

Flow-Based IDS Concept

Flow-based IDS uses **aggregated packet header information** instead of payload data. This approach reduces privacy concerns and improves efficiency when analyzing encrypted traffic.

Proposed Architecture

The proposed system consists of three major components:

1. Data acquisition and feature extraction
2. Data preprocessing
3. Decision engine (machine learning classifier)

The **system architecture diagram on page 3** illustrates the flow of data from datasets through preprocessing, training, and classification stages.

Ensemble Learning Method

The study evaluates several ensemble learning techniques, including:

- Adaptive Boosting (AdaBoost)
- Bootstrap Aggregation (Bagging)
- Random Forest
- Majority Voting

These techniques combine predictions from multiple classifiers to improve accuracy.

Dataset Used

The experiments were conducted using the **CIDDS-001 dataset**, which contains modern network traffic flows generated from a simulated business environment.

This dataset provides realistic traffic patterns and attack scenarios, making it suitable for evaluating intrusion detection systems.

Advantages of Ensemble Learning

Ensemble learning improves IDS performance by:

- Combining multiple weak classifiers
- Reducing overfitting
- Increasing classification accuracy
- Improving robustness of the detection system

Results

The experimental results indicate that **decision-tree-based ensemble classifiers outperform probabilistic and non-probabilistic methods** in terms of detection accuracy and training efficiency.

Conclusion

The research demonstrates that **ensemble machine learning methods significantly enhance intrusion detection performance**. Flow-based IDS combined with ensemble classifiers provides a scalable and efficient solution for detecting network attacks.

3. Literature Survey

Real-Time Intrusion Detection System Using Scapy with Hybrid Machine and Deep Learning Models

Paper:

Real-Time Intrusion Detection System Using Scapy with Hybrid Machine and Deep Learning Models

Objective

The paper proposes a **real-time intrusion detection system that integrates machine learning and deep learning models** for detecting cyber threats as they occur.

Proposed Method

The system combines two models:

1. **Autoencoder (Deep Learning)** – for anomaly detection
2. **K-Nearest Neighbor (KNN)** – for attack classification

The Autoencoder learns the normal behavior of network traffic, while KNN identifies the specific type of attack.

Dataset

The system is trained using the **NSL-KDD dataset**, which includes attack categories such as:

- DoS
- Probe
- Remote-to-Local (R2L)
- User-to-Root (U2R)

Real-Time Monitoring

The system captures live network packets using **PyShark**, which is a Python wrapper around TShark. These packets are converted into feature vectors and processed by the detection models.

Detection Process

The detection workflow includes:

1. Packet capture from the network interface
2. Feature extraction
3. Autoencoder anomaly detection
4. KNN attack classification
5. Alert generation

If suspicious activity is detected, the system sends **email alerts to administrators**.

Advantages

- Real-time network monitoring
- Detection of unknown attacks
- Automatic alerting mechanism
- Lightweight deployment

Conclusion

The hybrid Autoencoder-KNN model provides a powerful intrusion detection framework capable of detecting both known and unknown cyber threats.

4. Literature Survey

Real-Time Network Intrusion Detection using Machine Learning Techniques

Paper:

Real-Time Network Intrusion Detection using Machine Learning Techniques

Objective

This research focuses on designing a **real-time intrusion detection system using machine learning algorithms** to identify anomalous network traffic.

Methodology

The system captures network packets using **packet sniffing techniques** and analyzes them using trained machine learning models.

Machine Learning Algorithms Used

The study evaluates several algorithms:

- Support Vector Machine (SVM)
- Naïve Bayes
- Random Forest
- Decision Tree

Dataset

The system is trained using the **NSL-KDD dataset**, which contains both normal traffic and various attack types.

Detection Process

1. Capture network packets using PyShark
2. Extract relevant packet features
3. Compare packet features with trained models
4. Classify packets as normal or malicious
5. Generate alerts for detected attacks

Results

The Random Forest model achieved the highest performance with:

- Accuracy: **0.9911**
- High recall and precision values

Conclusion

Machine learning algorithms, particularly Random Forest, can effectively detect network intrusions in real time.

5. Literature Survey

Random Forest Based Intrusion Detection System

Paper:

Random Forest Based Intrusion Detection System

Objective

The paper proposes an intrusion detection system based on the **Random Forest algorithm** to improve detection accuracy and reliability.

Random Forest Algorithm

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions through voting.

Feature Selection

Feature selection is performed using:

- Pearson correlation analysis
- Internal feature importance evaluation

These techniques help identify the most relevant features for intrusion detection.

Advantages of Random Forest

- Handles high-dimensional data
- Reduces overfitting
- Provides feature importance scores
- Supports parallel processing

Results

The Random Forest IDS demonstrated improved performance in terms of:

- Accuracy
- Precision
- Recall

However, the study notes that further research is required to compare the method with other IDS algorithms.

Conclusion

Random Forest is an effective algorithm for intrusion detection due to its robustness and ability to process complex network data.